

“AI & IoT for Smart Energy Management in Green Systems”

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Abstract

The world is rapidly shifting toward cleaner, more sustainable energy solutions—and with good reason. As climate concerns grow and energy demands increase, traditional power systems are struggling to keep up. What we need now are energy infrastructures that are not only green but also smart able to adapt, self-optimize, and function efficiently under dynamic conditions [1]. This is where the powerful combination of Artificial Intelligence (AI) and the Internet of Things (IoT) comes into play.

AI and IoT together offer a transformative approach to managing the complex, decentralized nature of modern renewable energy systems. IoT devices such as smart meters, embedded sensors, and intelligent controllers allow us to monitor energy use, generation, and environmental conditions in real time [4]. These devices form a connected web that collects a continuous stream of data using lightweight communication protocols like LoRaWAN, ZigBee, and NB-IoT . Once gathered, this data is processed either at the edge or in the cloud to generate meaningful insights.

That’s where AI steps in. With techniques like machine learning, deep learning, and reinforcement learning, AI can analyze this massive data flow to predict energy demand, detect faults before they become failures, balance supply and demand, optimize loads, and even manage distributed power sources [7].

In this chapter, we dive into how AI and IoT technologies are architected to work together within energy systems. We break down the system layers, discuss key computational models, and explore their application in real-world contexts like smart grids, micro-grids, intelligent buildings, and smart factories. Through real-life case studies such as AI-powered solar micro-grids and IoT-driven wind turbine maintenance we showcase how these innovations are already reshaping energy landscapes around the world.

Of course, there are challenges: from cybersecurity and device compatibility to latency issues and data limitations [16]. But this evolving field is also full of promise. We explore exciting research directions such as federated learning for privacy-preserving AI, block chain for transparent energy trading, and digital twins for simulating energy systems [18].

Ultimately, this chapter lays the groundwork for building intelligent, reliable, and sustainable energy systems powered by AI, connected by IoT, and designed for a greener tomorrow.

1. Introduction

As the world’s energy needs continue to grow and the effects of climate change become increasingly urgent, our energy systems are facing a critical turning point. The traditional,

centralized grids powered by fossil fuels are giving way to a new generation of sustainable, intelligent, and resilient energy networks. At the heart of this transformation lies smart energy management—an approach that uses digital innovation to optimize how we generate, distribute, and consume energy.

Smart Energy Management Systems (SEMS) are crucial to this shift. They help reduce operational costs, boost energy efficiency, and seamlessly integrate renewable sources like solar and wind into the grid—all while minimizing environmental impact. However, managing the unpredictable nature of renewable energy, responding to real-time energy demands, and keeping the grid stable present major challenges. Traditional systems simply aren't equipped to handle this level of complexity.

That's where Artificial Intelligence (AI) and the Internet of Things (IoT) come in. IoT enables constant monitoring and data collection from connected devices like smart meters, sensors, and actuators deployed across the energy landscape. AI then analyzes this rich stream of real-time data, learning patterns, detecting anomalies, predicting future energy demands, and making intelligent decisions on the fly. Together, they pave the way for smart energy systems that are not only automated but also predictive, adaptive, and self-optimizing.

In this chapter, we dive deep into how AI and IoT work together to power smart energy management systems in today's green technology landscape. We explore technical architectures, enabling technologies, and real-world applications in smart grids, micro-grids, and energy-aware buildings. Case studies will illustrate how AI-IoT systems are already revolutionizing areas like demand-side management, predictive maintenance, and renewable energy integration.

We also tackle the challenges that come with these advancements—such as ensuring data security, managing communication latency, achieving device interoperability, and scaling solutions across large infrastructures. Finally, we look ahead to emerging frontiers: edge computing for faster on-site analytics, block chain for decentralized energy trading, digital twins for virtual simulations, and federated learning for privacy-preserving AI.

Whether you're a researcher, engineer, policymaker, or energy professional, this chapter offers a practical and forward-looking perspective on how AI and IoT can help build the intelligent, sustainable energy systems our future demands.

2. Understanding Smart Energy in Green Systems

2.1 What Are Green Systems?

Green systems also known as sustainable or eco-friendly systems—are built with the environment in mind. Their goal is simple yet powerful: reduce harm to the planet while making smarter, more efficient use of natural resources, especially energy. These systems play a crucial role in the global fight against climate change, helping cut down greenhouse gas (GHG) emissions and driving the shift toward a low-carbon economy.

What makes green systems stand out is their ability to function in balance with nature. They're designed not just for performance, but with a deep focus on sustainability. That means using less energy, reducing pollution, maximizing resource efficiency, and making renewable energy a core part of the mix. Importantly, these systems align with global Sustainable Development Goals (SDGs), particularly goals focused on clean energy (SDG 7), climate action (SDG 13), and building sustainable cities and communities (SDG 11).

Unlike traditional systems that depend on fossil fuels and centralized power structures, green systems embrace a different philosophy. They promote local energy generation, flexible demand management, and energy flows that support reuse and circularity—all to create a cleaner, more resilient future.

Key components of green systems include:

- **Renewable Energy Sources:** Solar photovoltaic (PV), wind, hydro, biomass, and geothermal systems serve as primary energy inputs, replacing or supplementing fossil fuels.
- **Energy-Efficient Infrastructure:** Smart buildings, green data centers, and optimized industrial processes that reduce energy consumption without compromising performance.
- **Distributed Energy Resources (DERs):** Small-scale, decentralized generation units such as rooftop solar panels, small wind turbines, and combined heat and power (CHP) units that operate closer to the point of consumption.
- **Energy Storage Solutions:** Battery energy storage systems (BESS), flywheels, super capacitors, and thermal storage units that enable load balancing and enhance grid reliability.
- **Electrified Transport and Smart Mobility:** Electric vehicles (EVs), electric buses, and integrated charging infrastructure that reduce transportation-related emissions.
- **Smart Grid Technologies:** Integration of digital tools such as advanced metering infrastructure (AMI), demand response systems, and communication protocols for real-time monitoring and control.

Today's green systems are getting smarter, thanks to the rapid rise of digital technologies. Tools like the Internet of Things (IoT) and Artificial Intelligence (AI) are no longer just add-ons—they're becoming essential. IoT enables real-time monitoring through a network of connected sensors, while AI brings powerful analytics that help systems learn, adapt, and improve over time. Together, they turn traditional green infrastructure into intelligent, responsive, and self-optimizing systems.

But green systems go far beyond just generating clean energy. They cover the entire energy journey from how power is produced and delivered, to how it's consumed, stored, and even recycled. This end-to-end perspective ensures energy is used responsibly, balancing environmental care with economic viability and social fairness.

As cities become smarter and industries move toward Industry 4.0, green systems will form the backbone of this transformation. Their role will be central in building a future where technology and sustainability go hand in hand.

2.2 Why Do We Need Smart Energy Management?

As modern energy systems become more complex—driven by the rise of distributed energy resources (DERs), the electrification of transportation, digital innovation, and the growing share of variable renewable energy (VRE)—traditional control methods are no longer enough. To keep up, we need smarter, faster, and more adaptive ways to manage energy.

This is where Smart Energy Management (SEM) comes in. SEM uses advanced technologies and data-driven strategies to keep energy systems running smoothly. Whether its generating power, storing it, distributing it efficiently, or making sure it's used wisely, SEM enables real-time monitoring and intelligent control. The result is a system that not only responds to change but also anticipates it—making energy use cleaner, more reliable, and more efficient.

1. Rising Energy Demand and Urbanization

As cities grow and electrification spreads across transport, heating, and cooling, global energy demand is set to rise sharply. Traditional grids, built for one-way power flow and centralized generation, struggle to keep pace with these shifting consumption patterns. Smart energy management offers a smarter alternative—using digital tools to align energy supply with real-time demand, particularly in urban environments where congestion and peak loads are a growing concern.

2. Integrating Renewable Energy Sources

Renewables like solar and wind are vital for cutting emissions, but their variability adds new challenges. Solar output dips during cloudy days, and wind is inherently unpredictable. Smart energy systems use forecasting tools and predictive analytics to anticipate these fluctuations. By adjusting storage or load in real time, they help maintain grid reliability and reduce energy waste from overproduction or curtailment.

3. Driving Energy Efficiency and Lower Costs

Smart energy systems can fine-tune how buildings and industrial plants consume power. For instance, production lines can be scheduled during off-peak hours to reduce costs, while HVAC and lighting systems can respond dynamically to occupancy and environmental conditions. These strategies lower utility bills and contribute to a more sustainable bottom line.

4. Embracing Decentralization and Modernizing the Grid

The energy landscape is shifting from centralized power stations to decentralized systems where consumers also produce energy—through rooftop solar, community micro-grids, or energy sharing platforms. Smart energy tools support this shift by enabling real-time communication and control across a network of distributed assets, boosting resilience and enabling local energy independence.

5. Supporting Smarter Demand Management

Demand-side management (DSM) helps reduce grid stress by shifting consumption away from peak hours. Smart systems automate DSM using real-time data, behavioral predictions, and price signals—nudging consumers to run appliances or charge electric vehicles when demand is lower. This reduces costs and supports grid stability.

6. Responding to Environmental and Regulatory Pressure

Governments are enforcing tougher energy efficiency standards and carbon regulations. Tools like emissions monitoring, automated reporting, and performance benchmarking—enabled by smart energy platforms—help companies stay compliant and track their sustainability progress.

7. Enabling Real-Time Monitoring and Maintenance

With IoT sensors and AI analytics, smart systems can monitor equipment performance continuously, detect issues early, and trigger maintenance before faults occur. This predictive approach minimizes downtime, lowers repair costs, and extends the life of critical infrastructure.

8. Empowering Consumers to Take Control

Smart energy dashboards and mobile apps give consumers insight into their usage habits and environmental impact. By visualizing real-time energy data, users can make more informed decisions—adjusting behavior, setting automation schedules, and actively participating in their energy ecosystem.

3. The Role of IoT in Smart Energy Management

The **Internet of Things (IoT)** is revolutionizing energy management by bringing intelligence, automation, and real-time connectivity to energy infrastructure. By embedding sensors and communication tools into physical assets—like meters, transformers, HVAC systems, solar panels, and home appliances—IoT enables these devices to share data, respond to changes, and operate more efficiently and sustainably.

3.1 IoT Architecture for Energy Systems

Smart energy systems powered by IoT typically have a four-layer architecture:

- **Perception Layer:** The foundation, made up of sensors and smart devices that monitor real-world variables such as energy consumption, temperature, occupancy, humidity, and sunlight levels.
- **Network Layer:** Transmits data from these sensors to centralized or edge computing systems using technologies like Wi-Fi, ZigBee, LoRaWAN, or NB-IoT.
- **Processing Layer:** Where raw data is turned into insights. Depending on needs, this might happen at the edge (for real-time response) or in the cloud (for deeper analysis).
- **Application Layer:** The user-facing component—dashboards, control panels, and alerts that help facility managers, homeowners, and operators make informed decisions.

3.2 Real-Time Monitoring and Energy Visibility

IoT devices provide a clear, live view of how energy is being used—down to individual appliances or zones in a building. Smart meters and sub-meters give granular insights that help identify inefficiencies, detect abnormal usage, and build load profiles for better planning.

For example, in a solar-powered home, sensors can monitor how much energy is being produced, how full the battery is, and how much energy the house is using. This helps decide whether to store, use, or sell the surplus power.

3.3 Forecasting Loads and Controlling Energy Use

Using historical data, weather forecasts, and occupancy patterns, machine learning models can predict energy usage with surprising accuracy. These insights allow proactive control strategies like:

- Pre-cooling buildings during cheaper off-peak hours.
- Shifting energy-heavy tasks to times when solar or wind power is most available.
- Avoiding energy spikes that trigger high utility charges.

In factories or plants, this also helps predict equipment downtime or schedule preventive maintenance, avoiding disruptions.

3.4 Smart HVAC and Lighting Control

IoT adds intelligence to heating, cooling, and lighting systems, which are major energy consumers in buildings:

- **Motion sensors** can turn off lights and adjust AC in empty rooms.
- **Smart thermostats** learn routines and optimize energy use for comfort and savings.
- **Light sensors** can reduce artificial lighting when there's enough natural daylight.

These features help reduce waste while maintaining user comfort.

3.5 Detecting Faults and Predicting Failures

IoT sensors continuously monitor machines for signs of trouble—unusual vibrations, rising temperatures, or energy spikes. When something's off, the system can flag it early, allowing maintenance teams to act before the problem worsens.

This kind of predictive maintenance lowers repair costs, avoids unexpected breakdowns, and improves equipment lifespan—especially vital in sensitive areas like hospitals or data centers.

3.6 Managing Renewables and Grid Interaction

IoT plays a vital role in managing Distributed Energy Resources (DERs) such as rooftop solar, wind turbines, and batteries. It helps by:

- Monitoring real-time renewable output and forecasting availability.
- Automatically charging or discharging batteries based on time-of-use prices.
- Coordinating with the grid to adjust loads in response to electricity pricing or supply constraints.

This two-way communication turns homes, buildings, and communities into active players in the energy market.

3.7 Smart Homes and Residential Energy Use

IoT is making smart homes even smarter when it comes to energy:

- Residents can control appliances, monitor solar output, and even schedule EV charging from their phones or voice assistants like Alexa or Google Assistant.
- Features like real-time usage tracking, utility bill forecasting, and carbon footprint estimates help promote sustainable behaviors.
- Smart scheduling can automate when devices run to avoid peak charges.

These systems empower homeowners to save money and reduce their environmental impact—all without sacrificing comfort.

3.8 Scalability and Interoperability

For IoT in energy to succeed on a large scale—think campuses, smart cities, or industrial zones—**interoperability** is key. Open protocols such as **MQTT, CoAP, BACnet, and Modbus** ensure that devices from different manufacturers can work together. Standards like **Open ADR** and **oneM2M** allow secure, seamless data exchange and future scalability.

4. How AI is Transforming Energy Systems

4.1 The Role of AI in Smart Energy Management

Artificial Intelligence (AI) is reshaping how energy systems operate—making them smarter, more efficient, and environmentally sustainable. When combined with the Internet of Things (IoT), AI helps energy systems go beyond basic automation. It introduces intelligent decision-making capabilities that allow systems to learn, predict, adapt, and optimize themselves over time. This shift is vital for managing energy more efficiently and supporting the transition to green energy.

4.1.1 Smarter Forecasting and Load Prediction

AI plays a critical role in predicting energy demand and renewable power generation. Using models like Neural Networks and Recurrent Neural Networks (RNNs), AI can analyze weather patterns, time-series data, and past consumption trends to accurately forecast:

- Daily or seasonal energy demand,
- Solar energy output based on cloud cover and irradiance,
- Wind power generation based on wind conditions.

These predictions help grid operators plan ahead, schedule generation efficiently, and make better use of battery storage—reducing waste and dependence on fossil fuels.

4.1.2 Dynamic Load Shifting and Demand Response

AI can help manage electricity usage more flexibly. Through intelligent demand response, it shifts energy use away from peak hours—like adjusting when appliances operate based on electricity prices or system needs. Reinforcement Learning (RL) agents are particularly effective at learning the best times to run heating, cooling, or charge electric vehicles, saving money and reducing grid stress without compromising user comfort.

4.1.3 Predictive Maintenance and Early Fault Detection

AI systems can spot early signs of equipment failure. Whether it's abnormal voltage levels, rising temperatures, or unusual vibrations, machine learning models detect these warning signs before breakdowns happen. Tools like Convolutional Neural Networks (CNNs) can even analyze thermal images, while other algorithms categorize faults and suggest preventive actions—helping avoid unexpected shutdowns and reducing maintenance costs.

4.1.4 Smart Optimization and Control

AI excels at solving complex optimization problems—like how to distribute energy most efficiently across different systems. Whether it's adjusting lighting, optimizing industrial machines, or balancing between solar panels and batteries, algorithms like Particle Swarm Optimization and Deep Reinforcement Learning help reduce costs, lower emissions, and maintain comfort or production levels.

4.1.5 Enhancing Grid Stability and Resilience

Renewable sources like wind and solar are unpredictable, and that can cause fluctuations in the grid. AI helps stabilize things by:

- Controlling voltage and frequency in real time,
- Reconfiguring microgrids during faults,
- Using intelligent systems to restore power faster.

AI also enables decentralized energy resources and virtual power plants to work together, maintaining grid stability even as more renewables come online.

4.1.6 Smarter Energy Markets and Trading

In deregulated markets, AI supports energy trading by:

- Forecasting prices and demand,
- Making strategic bids,
- Enabling peer-to-peer (P2P) energy exchanges using blockchain and smart contracts.

These systems allow households or businesses to trade energy like assets—boosting energy independence and fairness.

4.1.7 Personalized Energy Recommendations

AI can also help users directly. Based on usage patterns, it can offer custom energy-saving tips or adjust settings automatically. Thanks to Natural Language Processing (NLP), AI chatbots can explain consumption trends and suggest practical ways to reduce energy bills or environmental impact.

4.1.8 Integration with Smart Cities

AI doesn't stop at buildings—it plays a key role in smart city management. It helps coordinate:

- EV charging schedules,
- City-wide energy demand,
- District-level heating and cooling.

By aligning energy use across transport, water, and power systems, cities can become greener and more efficient.

4.1.9 Key IoT Applications in Energy

- **Smart Grids:** Monitor grid health and balance power distribution in real time.
- **Smart Homes:** Automatically switch off unused devices.
- **Demand Response:** Shift usage to off-peak periods.
- **Renewable Forecasting:** Align energy generation with weather patterns.

4.1.10 Real-World Examples

- **Wind Farms:** AI helps predict wind speeds, allowing for more reliable power generation.

- **Smart Factories:** AI schedules heavy equipment usage during off-peak hours to cut costs.

5. Integrated AI-IoT Framework for Smart Energy Systems

5.1 How AI and IoT Work Together

The synergy between Artificial Intelligence (AI) and the Internet of Things (IoT) creates a powerful ecosystem for smart energy management. In this setup:

- **IoT acts as the “senses”**—collecting data from the environment.
- **AI acts as the “brain”**—analyzing that data, making decisions, and taking action.

Together, they create systems that are aware, adaptive, and autonomous.

5.1.1 IoT: The Data Collection Backbone

IoT devices—like smart meters, sensors, and embedded controllers—track everything from:

- Energy consumption and power quality,
- Solar panel output and wind speed,
- Battery health and electric vehicle status,
- Environmental conditions like temperature and humidity.

These devices communicate via networks like Zigbee, LoRaWAN, or 5G to transmit real-time data to cloud or edge platforms for further processing.

5.1.2 AI: The Intelligent Decision Engine

Once data is collected, AI takes over. With tools like machine learning, fuzzy logic, and neural networks, AI can:

- Predict future consumption or solar output,
- Detect unusual patterns in equipment behavior,
- Optimize when and how devices operate.

AI makes sense of raw data and turns it into actionable insights or automated decisions.

5.1.3 Edge AI: Local and Fast

Edge AI moves some of the intelligence closer to the source—on local devices or gateways. This enables:

- Quicker decisions,
- Greater privacy,
- Better performance in places with poor connectivity.

For example, an AI-powered HVAC system can immediately adjust room temperature based on occupancy—without needing to consult a remote server.

5.1.4 Closed-Loop Control with Feedback

AI-IoT systems form a continuous loop of learning and improvement:

1. **Sensors Gather Data:** IoT devices monitor system conditions.
2. **AI Analyzes and Decides:** The system predicts demand, detects anomalies, or schedules resources.
3. **Actions Are Taken:** Devices respond by turning on, off, or adjusting output.
4. **System Learns and Adapts:** Based on feedback, the AI model improves its performance over time.

This loop helps maintain efficiency and reliability—even as conditions change.

Real-Life Applications of the Closed-Loop Approach

Example: Smart HVAC System in a Building

IoT sensors track room conditions and usage patterns. AI predicts when and where cooling or heating is needed, adjusting the system to reduce energy costs without sacrificing comfort.

Example: Solar Micro-grid with Battery Storage

IoT tracks energy generation, battery status, and demand. AI forecasts cloudy days or high usage hours, optimizing when to charge or discharge the battery. Feedback from each cycle helps the AI learn and improve over time.

Why Edge AI Matters in Energy Systems

For time-sensitive applications, edge AI is essential. It allows:

- Real-time responses,
- Increased security (no need to send data to the cloud),
- Reliable control even in unstable networks.

Mathematical Representation

In control theory terms, the system can be modeled using the following general structure:

- Let the system output be $y(t)$, reference value be $r(t)$, and control input be $u(t)$.
- The error signal is defined as $e(t)=r(t)-y(t)$.

- The AI controller generates $u(t)=f(e(t),x(t))$, where $x(t)$ represents contextual variables (e.g., time, weather, historical usage).
- Feedback from $y(t)$ is used to iteratively adjust f using a learning function L such that $f_{t+1}=L(f_t,e(t),x(t))$. This continuous feedback improves the AI controller's effectiveness over time.

5.1.5 Real-Time Visualization and Energy Dashboards

By combining IoT's real-time data collection with AI-driven analytics, energy dashboards can be created to visualize:

- Live energy usage and cost analytics
- Predictive trends and alerts
- Carbon footprint estimation
- Device-wise consumption breakdown
- Recommendations for energy-saving actions

These dashboards empower users, facility managers, and grid operators to make informed decisions and monitor system health.

5.1.6 Smart Micro-grids

Smart micro-grids represent a transformative evolution in modern energy systems, especially in the context of decentralized, resilient, and sustainable power distribution. Micro-grids are localized energy networks that can operate autonomously or in coordination with the main utility grid. The integration of Artificial Intelligence (AI) **and** Internet of Things (IoT) technologies into micro-grids has revolutionized how energy is generated, distributed, stored, and consumed, making these systems highly intelligent, adaptive, and self-healing.

Key Components of a Smart Micro-grid

A typical smart micro-grid consists of the following components:

- **Distributed Energy Resources (DERs):** Such as solar photovoltaic (PV) systems, wind turbines, micro-hydro units, and combined heat and power (CHP) plants.
- **Energy Storage Systems (ESS):** Including batteries, super capacitors, and flywheels.
- **Controllable Loads:** Smart appliances, electric vehicles (EVs), HVAC systems, and lighting systems.
- **IoT-Enabled Infrastructure:** Smart meters, sensors, relays, and controllers for real-time monitoring and control.
- **AI-Based Control Systems:** Algorithms for energy forecasting, load optimization, anomaly detection, and automated decision-making.

Role of IoT in Micro-grids

IoT acts as the **nervous system** of the micro-grid by ensuring seamless communication and coordination among all components. Real-time data is collected from DERs, loads, and storage systems using IoT devices like:

- Smart meters to track energy consumption
- Environmental sensors to assess weather conditions
- Battery management systems to report SoC and SoH
- Grid interface devices for import/export status

This data is transmitted via wireless (e.g., Zigbee, LTE-M, LoRa) or wired networks to edge gateways or cloud servers, where it becomes actionable intelligence.

Role of AI in Micro-grids

AI acts as the **brain** of the smart micro-grid, performing the following key functions:

- **Energy Demand Forecasting:** Predicts short- and long-term load profiles using machine learning models (e.g., LSTM, Random Forest).
- **Renewable Generation Prediction:** Estimates PV or wind output using weather forecasts, irradiance patterns, and historical data.
- **Optimal Dispatching:** Determines when to draw power from the grid, utilize DERs, or charge/discharge batteries to minimize cost and emissions.
- **Fault Detection and Diagnosis:** Identifies anomalies (e.g., inverter failures, grid faults) using AI classifiers and triggers corrective actions.
- **Islanded vs Grid-Connected Operation:** AI determines when the micro-grid should disconnect from the main grid (e.g., during outages) and operate in island mode.

Operational Flow: AI-IoT-Enabled Smart Micro-grid

1. **Monitoring:** IoT sensors collect real-time data from all sources.
2. **Analysis:** AI algorithms process the data to identify usage trends, generation capacity, and potential anomalies.
3. **Decision Making:** Based on forecasts and predefined objectives (e.g., cost reduction, carbon minimization), AI generates control strategies.
4. **Control Execution:** Commands are sent via IoT actuators to control switches, inverters, battery interfaces, or load devices.
5. **Feedback Loop:** System responses are continuously monitored to refine AI models through adaptive learning.

Technical Benefits

- **Increased Reliability:** Ability to maintain power supply during main grid outages through autonomous islanding.
- **Energy Cost Savings:** Peak shaving and demand-side management lower utility bills.

- **Improved Power Quality:** Voltage and frequency regulation with intelligent inverter control.
- **Carbon Footprint Reduction:** Optimal use of renewables and energy efficiency strategies.
- **Scalability and Modularity:** Micro-grids can expand or reconfigure with minimal disruption.

Real-World Application Example

In rural India, AI-IoT micro-grids are being deployed in villages with limited grid connectivity. IoT-based solar charge controllers track PV output, while AI models forecast evening demand spikes and adjust battery storage and diesel generator usage accordingly. The system dynamically prioritizes essential services like water pumps and clinics, ensuring uninterrupted operation even in remote locations.

Cybersecurity and Resilience Considerations

While the AI-IoT ecosystem enhances functionality, it also increases the system's **attack surface**. Smart micro-grids must implement robust security frameworks including:

- **Authentication and Encryption:** For secure data transmission.
- **Intrusion Detection Systems (IDS):** AI-based systems that detect abnormal behaviors or cyber threats.
- **Redundancy and Fail-Safe Mechanisms:** To maintain operation during hardware failures or communication loss.

5.1.7 AIoT in Smart Homes and Cities

In smart homes, IoT-based smart plugs, thermostats, and appliances work in conjunction with AI assistants (e.g., Alexa, Google Assistant) to learn user behaviors and reduce energy consumption through automation. At the city scale, AIoT platforms integrate data from traffic, lighting, and energy grids to optimize urban energy flows, manage EV charging, and ensure sustainable growth.

The convergence of Artificial Intelligence (AI) and the Internet of Things (IoT), commonly referred to as AIoT, is driving a paradigm shift in how energy is produced, consumed, and optimized in modern smart homes and cities. By embedding intelligence into everyday systems through sensors, actuators, and learning algorithms, AIoT enables context-aware, real-time decision-making that is pivotal for energy efficiency, sustainability, and user-centric design.

Smart homes equipped with AIoT technologies go beyond remote control and automation—they adapt and optimize energy consumption based on learned behaviors, environmental conditions, and real-time utility pricing.

1. Energy Monitoring and Optimization

IoT devices such as smart plugs, thermostats, and energy meters collect granular data on appliance usage, temperature, humidity, and occupancy. AI models analyze this data to:

- Predict energy demand patterns.
- Detect inefficient appliances or abnormal usage.
- Optimize HVAC, lighting, and water heating schedules.
- Schedule high-energy loads (e.g., EV charging, laundry) during off-peak hours.

2. Behavioral Learning and Personalization

AI algorithms learn user preferences over time to deliver personalized energy profiles. For instance:

- Adjusting thermostat settings based on historical comfort levels and weather forecasts.
- Automatically shutting down idle devices when the user leaves home.
- Recommending energy-saving actions through mobile apps or voice assistants.

3. Integration with Renewable Sources and Storage

AIoT systems manage **home-based solar PV systems** and **battery storage units**. The AI component forecasts solar generation and consumption, deciding when to store energy, draw from the grid, or export excess power—maximizing self-consumption and cost savings.

4. Smart Appliances and Edge Computing

New-generation appliances are embedded with edge AI chips that allow them to process data locally, reducing latency and enabling faster responses. This includes:

- Smart refrigerators adjusting internal temperatures based on usage.
- Washing machines that schedule cycles based on dynamic electricity pricing.

AIoT in Smart Cities

At the urban scale, AIoT is central to **Smart City** initiatives that aim to enhance quality of life, reduce environmental impact, and improve energy resilience. By integrating buildings, transport, public lighting, and utilities into a unified digital platform, municipalities can make cities more livable and sustainable.

1. Urban Energy Management Systems (UEMS)

City-wide energy flows are monitored using IoT-enabled smart meters, distribution sensors, and renewable energy installations. AI tools:

- Forecast city-scale energy demands.
- Optimize energy distribution from centralized and distributed sources.
- Balance supply-demand through dynamic grid reconfiguration.
- Detect and respond to outages or overloads in real time.

2. Smart Street Lighting

Smart lighting systems equipped with motion sensors and ambient light detectors adjust brightness dynamically, reducing energy consumption by up to 70%. AI algorithms optimize lighting schedules based on traffic patterns and event calendars.

3. EV Charging Infrastructure

AIoT enables intelligent scheduling and load balancing across a network of public EV chargers. The system can:

- Predict peak EV charging demand.
- Suggest optimal charging times and locations to users.
- Dynamically adjust charging rates to minimize grid stress.

4. Building Energy Management (BEMS)

Large commercial or residential buildings use AIoT for real-time monitoring of HVAC, elevators, lighting, and access control systems. These buildings integrate with city energy platforms to participate in demand response programs and peak load shaving.

5. Public Transportation and Traffic Management

IoT-connected traffic lights, buses, and smart parking meters generate data for AI models to optimize traffic flow and reduce idle times, leading to lower fuel usage and emissions. Smart transport systems are integrated with energy platforms to support electrification and charging infrastructure planning.

Technical Architecture

A typical AIoT architecture in smart cities and homes involves:

- **Sensing Layer:** IoT devices (sensors, meters, cameras).
- **Network Layer:** Communication protocols (Wi-Fi, Zigbee, NB-IoT, 5G).
- **Edge Layer:** On-device AI for local decisions.
- **Cloud/Platform Layer:** Centralized AI for coordination, data storage, and analytics.
- **Application Layer:** Interfaces for utilities, consumers, and city administrators.

Security and privacy are addressed via secure communication protocols, AI-based anomaly detection, and federated learning techniques to ensure data sovereignty.

Benefits of AIoT in Smart Homes and Cities

Aspect	Benefit
Energy Efficiency	Reduction in wastage, optimized appliance usage
Cost Savings	Lower utility bills through dynamic pricing and demand-side response
Environmental Impact	Reduced emissions via intelligent control and renewable integration
User Experience	Personalized automation, comfort, and convenience
Grid Resilience	Decentralized and distributed energy optimization
Data-Driven Governance	Real-time insights for policy makers and city planners

Real-World Implementations

- **Amsterdam Smart City (Netherlands):** AIoT-based systems manage energy flows, optimize traffic, and run pilot projects for smart buildings and zero-emission zones.
- **Barcelona (Spain):** Deployed a city-wide IoT platform that controls lighting, irrigation, and building energy through centralized AI coordination.
- **Tesla Smart Homes (USA):** Integration of AI, IoT, solar roofs, Power wall storage, and smart HVAC under a unified home energy management ecosystem.

5.2 Smart Office Building

Smart office buildings represent a transformative application of Artificial Intelligence of Things (AIoT) technologies, leveraging interconnected systems, real-time data, and machine learning algorithms to optimize energy efficiency, user comfort, and operational intelligence. As commercial real estate accounts for a significant portion of urban energy consumption particularly for lighting, HVAC (Heating, Ventilation, and Air Conditioning), and plug loads smart office infrastructure provides a strategic opportunity to achieve measurable sustainability gains in green systems.

Key Objectives of Smart Office Energy Management

- **Energy Efficiency:** Reduce unnecessary energy consumption without compromising occupant comfort.
- **Predictive Maintenance:** Avoid energy losses due to equipment degradation.

- **Occupant-Centric Comfort:** Align temperature, lighting, and ventilation based on occupancy and preferences.
- **Sustainable Operations:** Enable integration with renewable energy sources and demand response programs.
- **Operational Cost Reduction:** Optimize power usage and extend the life of mechanical systems.

5.2.1 Core AIoT Components in Smart Office Buildings

Component	Functionality
Smart Sensors	Monitor ambient temperature, humidity, light levels, motion, air quality.
AI-based Energy Management Systems (EMS)	Analyze energy data, forecast usage, automate control strategies.
Smart HVAC Systems	Dynamically adjust airflow and temperature based on occupancy and weather.
IoT-Enabled Lighting	LED lighting systems with presence detection, daylight harvesting, and dimming.
Occupancy Detection Systems	Use vision sensors or RFID to determine real-time occupancy and adjust systems.
Plug Load Controllers	Detect and manage non-essential electronics, shutting them off during inactivity.
Building Information Modeling (BIM) + Digital Twins	Real-time simulation and decision support using AI models.

5.2.2 AI and Machine Learning in Office Energy Management

AI enhances building management systems (BMS) through:

- **Pattern Recognition:** AI algorithms analyze historical data to identify usage trends and seasonal variability.
- **Predictive Analytics:** Forecasting energy demand and potential system failures based on real-time and past data.
- **Adaptive Control:** AI adjusts lighting and HVAC dynamically in response to environmental and behavioral factors.

- **Anomaly Detection:** Identifies abnormal energy use (e.g., equipment left on during off-hours or malfunctioning HVAC units).

Example: A neural network trained on HVAC usage data can predict the optimal setpoint temperatures for varying external conditions and occupancy levels, balancing energy consumption and comfort.

5.2.3 IoT Architecture for Smart Office Implementation

A layered approach is commonly adopted:

- **Perception Layer:** Sensors and actuators collect data (temperature, CO₂, light, noise, motion).
- **Network Layer:** IoT communication protocols such as ZigBee, LoRa, Wi-Fi, or BACnet over IP.
- **Edge Layer:** Local processing of sensor data using embedded AI for real-time response.
- **Cloud Layer:** Centralized analysis, data fusion, optimization algorithms, and visualization dashboards.
- **Application Layer:** Interfaces for facility managers, energy operators, and users to interact with the system.

5.2.4 Renewable Energy Integration and Energy Storage

Smart office buildings increasingly incorporate on-site renewable energy sources (e.g., rooftop PV panels) and Battery Energy Storage Systems (BESS). AI manages these systems by:

- Forecasting solar production using weather data and historical trends.
- Scheduling high-energy tasks when generation is high.
- Storing excess energy for peak load periods or grid outages.
- Enabling participation in demand response (DR) and virtual power plant (VPP) programs.

5.2.5 Human-Centric and Sustainable Design

Smart buildings enhance occupant productivity and well-being by dynamically adjusting indoor environmental quality (IEQ) parameters:

- **Thermal Comfort:** Personalized thermal zoning.
- **Visual Comfort:** Adaptive lighting that mimics natural daylight cycles.
- **Acoustic Management:** Active noise cancellation and acoustic zoning.
- **Air Quality:** Real-time air purification and CO₂ ventilation control.

These features contribute not only to energy savings but also to **employee satisfaction, cognitive function, and productivity**, aligning with WELL and LEED certification goals.

5.2.6 Real-World Examples

- **The Edge, Amsterdam:** A fully AI-integrated building using over 28,000 IoT sensors. It automatically adjusts light, HVAC, and space allocation while monitoring real-time occupancy.
- **Microsoft Smart Campus, Redmond:** Uses AI to monitor and control 125+ buildings for energy efficiency, saving millions in operational costs annually.
- **Infosys Bangalore:** Achieved over 40% reduction in energy usage by implementing AIoT-based building management and solar integration.

5.2.7 Benefits and Measurable Outcomes

Metric	Typical Improvement with Smart Office Systems
Energy Efficiency	20–40% reduction in overall energy consumption
Operational Cost Savings	25–35% drop in energy bills and maintenance costs
Carbon Emission Reduction	Up to 30% decrease in CO ₂ footprint
Occupant Satisfaction	Increased comfort, productivity, and indoor environment quality
System Uptime and Reliability	Predictive maintenance reduces downtime by over 50%

6. Case Studies

6.1 Solar Micro-grid in Rural India

Rural electrification in India remains a critical component of the country's energy access and sustainability agenda. In remote and underserved regions where grid extension is economically unviable or technically challenging, **solar micro-grids** provide a scalable, clean, and decentralized solution. Integrating **IoT and AI technologies** into these micro-grids enhances reliability, optimizes energy use, and ensures financial viability, creating a replicable model for sustainable development.

6.1.1 Background and Context

A case study conducted in the district of **Barabanki, Uttar Pradesh**, focuses on a village previously underserved by the national electricity grid. A **25 kW capacity solar photovoltaic (PV) micro-grid** was installed, designed to serve approximately 80 households, a school, and small commercial shops. The micro-grid included:

- Solar panels,

- Inverter and charge controller,
- Lithium-ion battery bank (120 kWh capacity),
- Smart meters,
- IoT sensors,
- AI-enabled energy management software.

6.1.2 Role of IoT in the Micro-grid

IoT devices played a foundational role in monitoring, control, and maintenance:

- **Smart Meters:** Installed at every consumer node, enabling real-time consumption tracking, remote disconnection, and pay-as-you-go (PAYG) billing.
- **Environmental Sensors:** Measured irradiance, temperature, and humidity to forecast solar power generation capacity.
- **Energy Flow Sensors:** Monitored voltage, current, and frequency at various points to detect faults or inefficiencies.

These sensors transmitted data via LoRaWAN and GSM networks to a cloud-based analytics platform.

6.1.3 AI-Driven Energy Management

Artificial Intelligence algorithms performed the following:

- **Load Forecasting:** Predicted daily and weekly load patterns using historical consumption, weather forecasts, and event data.
- **Generation Forecasting:** Estimated solar energy production based on solar irradiance, panel orientation, and atmospheric conditions.
- **Battery Optimization:** AI optimized charge/discharge cycles to extend battery life and minimize depth-of-discharge (DoD).
- **Demand-Side Management (DSM):** AI dynamically scheduled appliance usage (e.g., irrigation pumps or grinders) to off-peak hours or when surplus energy was available.

6.1.4 User Engagement and Mobile Interface

A custom Android mobile application was deployed:

- Allowed users to check their energy balance, usage, and pricing.
- Enabled remote load control (e.g., turn off lights or fans when not in use).
- Provided alerts for maintenance, battery health, and load curtailment.

The mobile platform contributed significantly to **energy literacy** and **user empowerment**.

6.1.5 Performance Metrics

6.1.6 Impact Assessment

Parameter	Before Micro-grid	After AI-IoT Solar Micro-grid
Electricity Availability	0–4 hours/day	24/7 for essential loads
Monthly Energy Access per HH	<5 kWh	20–25 kWh
System Downtime	Frequent, unreported	<3% with AI-based fault prediction
Billing & Revenue Collection	Manual, 60% collection efficiency	Digital PAYG, 97% efficiency
CO ₂ Emissions	High (diesel use)	Reduced by 15–20 tons/year

The deployment of an AIoT-enabled solar micro-grid delivered substantial socio-economic and environmental benefits:

- **Improved Education:** Evening study hours increased by 3–4 hours/day.
- **Entrepreneurship:** Shops and micro-enterprises extended operational hours.
- **Healthcare Access:** Reliable electricity for vaccine refrigeration and basic diagnostics.
- **Gender Inclusion:** Women-led home businesses (e.g., tailoring) gained productivity.
- **Environmental Impact:** Diesel generator use reduced by 90%.

6.1.7 Scalability and Reliability

The success of this project illustrates a **scalable and modular framework** for rural energy access:

- Modular inverter + PV + battery designs.
- AI/IoT platform adaptable to local grid conditions.
- PAYG model supports financial sustainability and user accountability.
- Can integrate with government schemes like Saubhagya and UJALA.

6.2 Smart Campus in Singapore

Singapore is internationally recognized for its commitment to digital innovation and sustainable urban development. A flagship example is the Smart Campus initiative by the National University of Singapore (NUS), which integrates AI and IoT technologies to transform the university into a living lab for intelligent energy and environmental management. The NUS Smart Campus exemplifies how integrated systems can deliver high-efficiency, data-driven decision-making for large, mixed-use facilities.

6.2.1 Overview of the Campus Energy Ecosystem

The NUS Smart Campus covers over **150 buildings**, serving 40,000+ students and faculty. It combines a **grid-connected energy network** with renewable sources, high-performance buildings, and centralized monitoring platforms.

Key energy infrastructure includes:

- 2.5 MWp of **rooftop solar PV systems**
- 1.2 MWh **battery energy storage system (BESS)**
- High-efficiency **chilled water district cooling systems**
- Smart metering and IoT-based environmental sensing
- AI-driven Energy Management System (EMS)

6.2.2 IoT-Driven Monitoring Infrastructure

The deployment of thousands of IoT devices enables real-time visibility into every layer of the campus's energy and environmental footprint. These include:

- **Smart Meters** at the device, floor, and building levels
- **Environmental Sensors** (CO₂, temperature, humidity, light levels)
- **Occupancy Sensors** (RFID, motion-based)
- **Water and Waste Monitoring** nodes

These devices use **NB-IoT**, **Zigbee**, and **Wi-Fi mesh networks** to communicate with a central data lake hosted on a secured cloud platform.

6.2.3 AI-Based Optimization and Automation

Artificial Intelligence models are embedded into the campus-wide EMS, leveraging historical and real-time data to optimize performance.

Key applications include:

- **Load Prediction and Scheduling:** AI forecasts 15-minute interval loads to optimize energy procurement and HVAC operations.
- **Adaptive Lighting Control:** Based on occupancy and daylight levels, AI adjusts luminaire intensity in real-time.
- **Smart HVAC Operation:** AI algorithms dynamically adjust air flow, temperature set-points, and chiller loads based on occupancy patterns and weather forecasts.
- **Anomaly Detection:** AI identifies abnormal energy use (e.g., water leaks, power surges) and sends real-time alerts for maintenance action.

These algorithms are built using time-series machine learning models such as LSTM (Long Short-Term Memory) and XGBoost regression trees.

6.2.4 Human-Centric Interfaces and User Engagement

User dashboards and mobile apps enhance engagement and sustainability behavior:

- Real-time **energy and carbon dashboards** on kiosks and apps
- Personalized **carbon footprint scores** for students and staff
- Gamified energy-saving challenges between departments
- Feedback loops through surveys and satisfaction scoring

This results in heightened environmental awareness and active participation in the sustainability mission.

6.2.5 Performance and Outcomes

The smart energy management ecosystem has delivered measurable benefits over 3 years:

Metric	Before (2018)	After Smart Campus (2023)
Energy Consumption	1.8 GWh/month	1.45 GWh/month (↓ ~20%)
HVAC Energy Use	50% of total load	38% of total load (↓ by ~25%)
Peak Load Reduction	—	12% via AI load shifting
Solar Utilization	40%	82% (with storage integration)
Building Occupant Satisfaction	Moderate	High (via smart comfort control)
Annual Carbon Emissions Reduction	—	~1,500 metric tons CO ₂ /year

6.2.6 Research and Innovation Hub

The NUS Smart Campus also serves as a **test bed for R&D** in urban sustainability:

- Digital twins for building energy simulation and retrofitting.
- Block chain-based energy trading pilots with neighboring campuses.
- Collaboration with startups on sensor design and AI integration.
- Use of edge AI in low-power embedded devices for decentralized processing.

This creates a vibrant ecosystem that nurtures innovation while simultaneously reducing operational costs and emissions.

6.2.7 Lessons Learned and Scalability

Key takeaways from the smart campus deployment include:

- **Data Standardization** is critical for interoperability between devices and platforms.
- **Edge-Cloud Hybrid Architecture** balances latency-sensitive control (e.g., lighting, HVAC) with centralized analytics.
- **User Involvement** is essential: behavior accounts for up to 30% of energy performance.
- **Scalability**: The same model is now being replicated across other institutions in Singapore, such as NTU and SMU.

7. Benefits of AI and IoT in Energy Systems

- **Energy Efficiency**: Reduces waste and improves resource usage
- **Cost Savings**: Cuts down on electricity bills and maintenance expenses
- **Sustainability**: Enables more effective use of renewable energy
- **Reliability**: Prevents equipment failures through early warnings
- **User Engagement**: Real-time dashboards and alerts increase awareness and control

8. Challenges and Limitations

While promising, some challenges still need to be addressed:

- **Data Security and Privacy**: Energy data can be sensitive; proper encryption and access control are essential.
- **Standardization**: Devices from different vendors may not be compatible.
- **Initial Costs**: Deployment of smart systems can be expensive.
- **Technical Skills**: Requires trained personnel to operate and maintain systems.
- **Scalability**: Managing millions of IoT devices and AI models in large grids can be complex.

9. Future Directions

The future of AI-IoT in energy management looks bright, with several emerging trends:

- **Edge AI**: Running AI models on local devices to reduce latency and improve speed.
- **Federated Learning**: Training AI models locally across multiple devices while keeping data private.
- **Blockchain Integration**: Enabling secure peer-to-peer energy trading.
- **Digital Twins**: Virtual models of physical energy systems that can simulate behavior and test improvements.

10. Conclusion

The future of energy is becoming smarter, greener, and more connected—thanks to the powerful combination of Artificial Intelligence (AI) and the Internet of Things (IoT). As we face growing energy demands and climate challenges, traditional systems are no longer enough. Instead, we need energy networks that can think, adapt, and act in real-time. That's exactly what AI and IoT enable.

Throughout this chapter, we explored how AI and IoT together bring intelligence to green energy systems. From predicting energy demand to managing solar and wind power, from detecting equipment faults before failure to balancing supply and demand, these technologies are changing the way we produce, store, and consume energy. Real-world examples—from solar micro-grids in rural India to smart campuses and wind farms—demonstrate how these innovations are already making a positive impact.

However, there are still challenges to overcome: securing data, making different systems work together, protecting user privacy, and creating supportive policies. To truly unlock the potential of smart energy, we need collaboration between researchers, governments, businesses, and communities.

As technology evolves, the next steps may include smarter edge devices, digital twins for energy systems, decentralized power sharing through block chain, and even more personalized energy usage insights. These innovations will help us move toward an energy future that is not just smart, but also sustainable, inclusive, and resilient.

References

1. Khan, M. A., Rehmani, M. H., & Reisslein, M. (2019). Smart grid communication network: Survey on the communication architectures, standards, and challenges. *Computer Networks*, 144, 295–318. <https://doi.org/10.1016/j.comnet.2018.07.020>
2. Bui, N., & Zorzi, M. (2018). Health care applications: A solution based on the Internet of Things. *Computer Communications*, 54, 78–88. <https://doi.org/10.1016/j.comcom.2014.08.010>
3. Mengelkamp, E., Gärttner, J., Rock, K., Kessler, S., Orsini, L., & Weinhardt, C. (2018). Designing microgrid energy markets: A case study: The Brooklyn Microgrid. *Applied Energy*, 210, 870–880. <https://doi.org/10.1016/j.apenergy.2017.06.054>
4. Zhang, Y., Deng, R., Yang, H., & Qian, Y. (2020). Blockchain-based distributed control and security architecture for intelligent distributed energy resources. *IEEE Transactions on Industrial Informatics*, 16(10), 6665–6674. <https://doi.org/10.1109/TII.2019.2941951>
5. Lu, Y., & Xu, X. (2019). Resource-aware distributed deep learning at the edge. *IEEE Network*, 33(6), 130–137. <https://doi.org/10.1109/MNET.001.1900082>

6. Shukla, A., Singh, P., & Banerjee, R. (2020). Performance evaluation of solar microgrid in India. *Renewable and Sustainable Energy Reviews*, 133, 110290. <https://doi.org/10.1016/j.rser.2020.110290>
7. Liu, Y., Wu, J., Long, C., & Thomas, L. (2019). Peer-to-peer energy sharing through a two-stage aggregated battery control in a community microgrid. *Applied Energy*, 226, 261–276. <https://doi.org/10.1016/j.apenergy.2018.04.134>
8. Ahmad, T., Chen, H., Wang, J., & Huang, C. (2020). Artificial intelligence in renewable energy systems: A review. *Renewable and Sustainable Energy Reviews*, 119, 109537. <https://doi.org/10.1016/j.rser.2019.109537>
9. Abdel-Basset, M., Mohamed, R., & Zhou, Y. (2019). Edge computing models, technologies and applications: A systematic review. *Future Generation Computer Systems*, 97, 219–240. <https://doi.org/10.1016/j.future.2019.02.007>
10. Cao, Y., Yang, H., & Wang, S. (2021). Digital twin and big data analytics for improving energy efficiency and operational performance of HVAC systems. *Energy and Buildings*, 224, 110267. <https://doi.org/10.1016/j.enbuild.2020.110267>
11. Zhang, C., Wu, J., Zhou, Y., Cheng, M., & Long, C. (2018). Peer-to-peer energy trading in a microgrid. *Applied Energy*, 220, 1–12. <https://doi.org/10.1016/j.apenergy.2018.03.010>
12. Kim, J., Shin, H., & Lee, J. (2022). Explainable AI for energy systems: Status, challenges, and future directions. *IEEE Access*, 10, 50289–50308. <https://doi.org/10.1109/ACCESS.2022.3172754>
13. Siano, P., De Maio, G., & Ranganathan, P. (2020). Real-time demand response and energy management using edge-AI in smart grids. *Sustainable Cities and Society*, 61, 102322. <https://doi.org/10.1016/j.scs.2020.102322>
14. Chakraborty, S., & Mohanpurkar, M. (2021). Federated learning for power systems: Privacy-preserving analytics. *Electric Power Systems Research*, 194, 107057. <https://doi.org/10.1016/j.epsr.2021.107057>
15. Jain, A., & Bansal, A. (2022). Predictive maintenance in wind turbines using AI and IoT. *Renewable Energy*, 189, 1306–1317. <https://doi.org/10.1016/j.renene.2022.02.012>
16. Uddin, M., & Rahman, M. (2021). IoT-driven smart grid for improving energy efficiency: A comprehensive review. *Energy Reports*, 7, 6049–6066. <https://doi.org/10.1016/j.egyr.2021.09.085>
17. Sharma, R., & Saxena, A. (2023). Role of AI in optimizing battery life and power balancing in solar microgrids. *Journal of Energy Storage*, 65, 107038. <https://doi.org/10.1016/j.est.2023.107038>

18. Gupta, V., Singh, H., & Arora, A. (2022). An intelligent IoT-enabled framework for real-time monitoring of renewable energy systems. *Computers & Electrical Engineering*, 102, 108215. <https://doi.org/10.1016/j.compeleceng.2022.108215>
19. Feng, Y., Zhang, L., & Hu, X. (2024). Secure AIoT in smart homes: A review of techniques and trends. *IEEE Internet of Things Journal*, 11(2), 1572–1589. <https://doi.org/10.1109/JIOT.2023.3325260>
20. Mishra, R., & Kumar, A. (2025). Blockchain-integrated peer-to-peer energy trading in microgrids. *Energy Informatics*, 8(1), 22. <https://doi.org/10.1186/s42162-025-00349-2>